# MATH 515—Applied Mathematics
# Syllabus—Fall, 2026

**Instructor:** Dr. Bryce Alan Christopherson
**Email:** bryce.christopherson@und.edu
**Class:** TR 2:00 PM–3:15 PM
**Office:** Witmer Hall, Room 321
**Location:** Witmer 309
**Office Hours:** MW 11:00 AM–12:00 PM

**Document Information:** This syllabus describes the requirements and procedures for MATH 515—Applied Mathematics. You are responsible for knowing this material, so please read it carefully. Substantive changes will be announced via Blackboard announcement. You will be responsible for any changes. Your continued enrollment in this course is your implicit agreement to abide by the requirements of this class.

**Prerequisites:** MATH 266 or an equivalent differential-equations course, together with working linear algebra. Students without the formal prerequisite may enroll with the instructor's consent.

**Course Description:** This is an introduction to nonlinear ordinary differential equations from an applied and geometric point of view. We will learn how to extract qualitative information when an explicit solution is unavailable: equilibria and their stability, bifurcations, phase portraits, invariant regions, conserved quantities, Lyapunov functions, and periodic orbits. Analytical work will be paired with reproducible numerical experiments in Python. This course is the first half of a two-semester sequence.

**Course Materials:** The materials for this course, both required and optional, are as follows:

**Textbook (Required):** Steven H. Strogatz, *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*, 3rd ed., CRC Press, 2024. The 3rd edition is the official reference edition, but the 2nd edition is fully adequate for this course. Page, figure, equation, and exercise numbers may differ between editions.

**Computing:** Google Colab with Python, NumPy, SciPy, and Matplotlib is the supported computational environment. No local installation is required. Other software is welcome, but students are responsible for supplying reproducible code and for translating platform-specific output when needed.

**Calculator (optional):** A scientific non-graphing calculator or computer equivalent is permitted, though not required, for this course.

**Blackboard:** All materials for this course will be posted to the class Blackboard page. Any updates or changes pertaining to the course will be posted as an announcement in Blackboard. Adjust your notification settings so that you receive emails or texts when announcements are posted. You want to be sure that you do not miss updates on assignments or other important information.
To access Blackboard, please go to: http://blackboard.UND.edu and log in with your NDUS Identifier (Username and Password). If you do not know your NDUS Identifier or have forgotten your password, please visit your NDUS account webpage on the University Information Technologies (UIT) website. For technical assistance, please contact UND Technical Support at 701.777.2222, visit the UIT website for their hours, help documents, and other resources, or visit the Knowledge Base for additional supports and information about general tech requirements for students including information about devices, operating systems, software, internet connection, and major-specific tech requirements.

**Grading Policy:** Your percentage grade is determined by the components detailed in the table below. You can estimate your letter grade by using the following scale: 90–100% is an A, 80–89% is a B, 70–79% is a C, 60–69% is a D, and 0–59% is an F. Plus/Minus Grades will not be awarded.

| | |
|---|---|
| Eight problem sets | 45% |
| Midterm 1 | 15% |
| Midterm 2 | 15% |
| Comprehensive final examination | 25% |

**Incompletes:** It is expected that students will complete all requirements for a course during the time frame of the course. For reasons beyond a student's control, and upon request by the student or on behalf of the student, an incomplete grade may be assigned by the instructor when there is reasonable certainty the student will successfully complete the course without retaking it. The mark "I," Incomplete, will be assigned only to the student who has been in attendance and has done satisfactory work up to a time within four weeks of the close of the semester, including the examination period, and whose work is incomplete for reasons satisfactory to his or her instructor. More information regarding UND's Incomplete policy can be found on the UND "Grading System" webpage.

**Exams:** All exams except the final exam will be administered during the regularly scheduled class time. On examinations, you will be given a number of problems (generally, similar to those assigned in the homework) to complete during a fixed time period. To receive full credit on your submission, you are **required** to show **all** of your work in a **neat**, **coherent**, and **mathematically correct** fashion. A make-up exam will only be administered if there is documentation from a proper authority, such as a note from a physician in the case of illness. If you know that you will be unable to complete an exam in the allotted time, please talk to your instructor at least one week before the date the exam will be administered. If you have any questions about the scheduling of the exams, please discuss this with your instructor.

**Make-up Exams:** If you will not be able to take an exam on the scheduled date and time you must make other arrangements with the instructor well in advance. In cases where proper notification has not been given, or if the reason for missing the exam is not valid, the make-up exam may be penalized or may not be given.

**Homework:** Eight homework assignments (equally weighted) will be given during the semester. Except where an assignment gives an explicit selection rule, all problems are expected. To assist with timely feedback, two analytical problems will be chosen for detailed grading after the deadline. The computational or modeling component designated on each assignment, together with the completeness and clarity of the remaining work, will also be graded. A complete solution key will be posted after the late-work window. All submitted work must be neat, coherent, legible, and sufficiently detailed to communicate your reasoning. Your work will be evaluated for mathematical correctness and the quality of your analysis and explanation.

Due dates for assignments will appear in the course calendar and on Blackboard. All times are listed in the Central Time Zone. There is an automatic 48-hour grace period after the due date to submit late work before the key is posted. After a key is posted, late assignments will not be accepted.

**Extra Credit:** Opportunities for earning extra credit may occur throughout the semester at the discretion of the instructor. Extra credit problems may be given in class or occasionally may be offered on exams. Points earned from extra credit may help borderline grades, but do not expect extra credit to significantly change your final score.

**Attendance:** Attending regularly scheduled lecture sessions and/or viewing lecture videos is not mandatory, but I believe it is very important. If you frequently do not attend/view the lectures without a good reason and begin to struggle in the course, I will have very little sympathy for you.

**Getting Help:** One of the most important skills a student can develop is the ability to recognize quickly when they are struggling with some concept in a class. Once you recognize you need help, first check through your materials (such as your textbook or class notes, for example) to see if the answer is there. If you cannot alleviate whatever difficulties you are having yourself, *please seek help*. There are lots of resources available to you. It is important that you try a variety of different services. If you do not receive the help you need at that time, please try it again or try a different service. Don't be shy—make sure you get the help you need!

**Study Groups:** Meet with some of your classmates and help each other with questions. Learning from your peers is one of the best ways to learn.

**Office Hours:** My office hours (subject to change as noted above) are listed at the top of this document. Please seek me outside of class if you have any questions, concerns, or would like any additional help with the course material. If these times are not convenient, we should be able to arrange an alternative time. I am here to help you!

**UND Services:** UND cares about your success as a student and provides a number of services, such as The UND Writing Center, Tutoring and Learning Services, Testing Services, and more. For additional information, visit the UND "Student Resources" webpage.

**Academic Integrity:** Academic integrity is a serious matter, and any deviations from appropriate behavior will be dealt with strongly. As a scholastic matter, the professor has the discretion to determine appropriate penalties for the student's workload or grade, but the situation may be resolved without involving many individuals. An alternative is to treat the situation as a disciplinary matter, which can result in suspension from the University, or have lesser penalties. Be aware that I view this as a very serious matter and will have little tolerance and/or sympathy for questionable practices. A student who attempts to obtain credit for work that is not their own (whether that be on a paper, quiz, homework assignment, exam, etc.) will likely receive a failing grade for that item of work, and at the professor's discretion, may also receive a failing grade in the course. For more information read the Code of Student Life.

**Disability Statement:** The University of North Dakota is committed to providing equal access to students with documented disabilities. To ensure access to this class and your program, please contact Student Disability Resources (SDR) to engage in a confidential discussion about accommodations. Accommodations are not provided retroactively. Students are encouraged to register with SDR at the start of their program. More information can be obtained by email (UND.sdr@UND.edu) or by phone at 701-777-2100.

**Course Evaluation:** Near the end of the semester, you will be asked to complete an online course evaluation form (SELFI). Your feedback on the course is extremely valuable to me. I read my students' comments carefully and use them to improve the course the next time I teach it. When the time comes, please let me know which aspects of the course helped you learn—and which aspects might be modified to help future students learn more effectively. Please note that the course evaluations are anonymous and that I won't see the results until after the grades for the course are submitted, allowing you to provide honest and constructive feedback. Throughout the semester if you have concerns or feedback, please reach out to schedule a time to discuss.

**UND Syllabi Statements:** Additional syllabi statements are provided by the Office of the Provost.

**Standards for Written Mathematical Work** The rubric below describes the general standards expected of written solutions in this course. It is intended as a guideline rather than as a mechanical point-allocation formula; individual problems may emphasize different aspects of the rubric. You should strive to present work that is clear and well organized, formulate the mathematical problem correctly, use notation and terminology precisely, justify the methods and calculations employed, and interpret the resulting mathematics in the context of the problem.

Because this is a graduate-level course, a correct final answer will not ordinarily constitute a complete solution by itself. A complete solution should make clear why the chosen method is appropriate, how the conclusion follows from the analysis, and what the result says about the system or model being studied. Please consult this rubric when preparing solutions, reviewing feedback, or asking questions about the evaluation of your work.

| **Characteristics of the Written Work** | **Qualitative Description** | **Typical Scoring %** |
|---|---|---|
| (i) The work is clear, well organized, complete, and easy to follow. The overall strategy is apparent.<br>(ii) Variables, parameters, assumptions, initial or boundary conditions, and other relevant quantities are properly identified. Mathematical notation and terminology are used precisely.<br>(iii) The chosen analytical or computational method is appropriate, and the derivation is logically justified. Relevant definitions, theorems, and hypotheses are used correctly.<br>(iv) Calculations are correct, and any graphs, phase lines, phase portraits, bifurcation diagrams, or numerical results are accurate, clearly labeled, and appropriately interpreted. The work may contain a minor typographical or arithmetic error that does not affect the essential analysis. | Exceptional | 90–100 |
| (i) The work is generally clear, organized, and essentially complete, although a few steps may be presented too briefly.<br>(ii) The mathematical formulation, assumptions, and notation are mostly correct, with only minor omissions or ambiguities.<br>(iii) The chosen method is appropriate and the main derivation is correct, although one justification, hypothesis, or computational step may not be fully explained.<br>(iv) Calculations and graphical or numerical work are essentially correct, and the conclusion is reasonably interpreted. Any gap or error can be repaired locally without changing the main strategy. | Good | 75–89 |
| (i) A relevant approach can be identified, but the work may be incomplete, poorly organized, or difficult to follow in places.<br>(ii) The mathematical formulation is only partially complete, or some variables, assumptions, conditions, or notation are left unclear.<br>(iii) The work contains a substantive gap, misapplication of a method, or computational error that requires nontrivial repair, although significant portions of the analysis are correct.<br>(iv) Graphical or numerical work may be partially correct but incomplete, insufficiently labeled, or inadequately connected to the analysis. The conclusion demonstrates partial understanding but is not fully justified. | Fair | 60–74 |
| (i) The work contains some relevant ideas, but the intended approach is incomplete or difficult to reconstruct.<br>(ii) The mathematical problem is formulated incorrectly or incompletely, and important assumptions, conditions, or quantities may be omitted.<br>(iii) The solution contains several major gaps, unjustified steps, inappropriate methods, or substantial computational errors.<br>(iv) Graphical or numerical work is largely incorrect or incomplete, and the conclusion is unsupported or incorrectly interpreted. The work demonstrates limited understanding and cannot be corrected by repairing one isolated step. | Poor | 40–59 |
| (i) The work presents no coherent strategy relevant to the problem or makes little meaningful progress.<br>(ii) There is little or no correct mathematical formulation, and the relevant variables, assumptions, or conditions are not identified.<br>(iii) Formulas or computational results are presented without meaningful justification, or the work is primarily irrelevant or contradictory.<br>(iv) Required graphs or numerical results are absent or fundamentally incorrect, and the response demonstrates little or no understanding of the mathematical ideas involved. | Very Poor | 0–39 |

**Course Schedule:** (Chapters Covered: 1-7)

| Week/Day | Su. | Mo. | Tu. | We. | Th. | Fr. | Sa. |
|---|---|---|---|---|---|---|---|
| Week | 23 Aug. | 24 Aug. | 25 Aug. | 26 Aug. | 27 Aug. | 28 Aug. | 29 Aug. |
| 1 | | | Intro<br>1.0-1.3 | | 2.0-2.2 | | |
| Week | 30 Aug. | 31 Aug. | 1 Sept. | 2 Sept. | 3 Sept. | 4 Sept. | 5 Sept. |
| 2 | | | 2.3-2.4 | | 2.5-2.6 | | |
| Week | 6 Sept. | 7 Sept. | 8 Sept. | 9 Sept. | 10 Sept. | 11 Sept. | 12 Sept. |
| 3 | | *Labor Day*<br>*(No Class)* | 2.7 | | 2.8 | **HW 1** | |
| Week | 13 Sept. | 14 Sept. | 15 Sept. | 16 Sept. | 17 Sept. | 18 Sept. | 19 Sept. |
| 4 | | | 3.0-3.1 | | 3.1 | | |
| Week | 20 Sept. | 21 Sept. | 22 Sept. | 23 Sept. | 24 Sept. | 25 Sept. | 26 Sept. |
| 5 | | | 3.2 | | 3.4 | **HW 2** | |
| Week | 27 Sept. | 28 Sept. | 29 Sept. | 30 Sept. | 1 Oct. | 2 Oct. | 3 Oct. |
| 6 | | | 3.4 | | 3.3,3.5 | **HW 3** | |
| Week | 4 Oct. | 5 Oct. | 6 Oct. | 7 Oct. | 8 Oct. | 9 Oct. | 10 Oct. |
| 7 | | | Review | | **Exam 1**<br>**(Ch. 1-3)** | | |
| Week | 11 Oct. | 12 Oct. | 13 Oct. | 14 Oct. | 15 Oct. | 16 Oct. | 17 Oct. |
| 8 | | | 4.0-4.3 | | 4.4-4.6 | | |

| Week/Day | Su. | Mo. | Tu. | We. | Th. | Fr. | Sa. |
|---|---|---|---|---|---|---|---|
| Week | 18 Oct. | 19 Oct. | 20 Oct. | 21 Oct. | 22 Oct. | 23 Oct. | 24 Oct. |
| 9 | | | 5.0-5.1 | | 5.2 (Part 1) | **HW 4** | |
| Week | 25 Oct. | 26 Oct. | 27 Oct. | 28 Oct. | 29 Oct. | 30 Oct. | 31 Oct. |
| 10 | | | 5.2 (Part 2), 5.3 | | 6.0-6.2 | | |
| Week | 1 Nov. | 2 Nov. | 3 Nov. | 4 Nov. | 5 Nov. | 6 Nov. | 7 Nov. |
| 11 | | | 6.3 | | 6.3, 8.1 | **HW 5** | |
| Week | 8 Nov. | 9 Nov. | 10 Nov. | 11 Nov. | 12 Nov. | 13 Nov. | 14 Nov. |
| 12 | | | 7.1-7.3 | *Veteran's Day (No Classes)* | 8.2, 8.7 | | |
| Week | 15 Nov. | 16 Nov. | 17 Nov. | 18 Nov. | 19 Nov. | 20 Nov. | 21 Nov. |
| 13 | | | Review | | **Exam 2 (Ch. 4-6)** | **HW 6** | |
| Week | 22 Nov. | 23 Nov. | 24 Nov. | 25 Nov. | 26 Nov. | 27 Nov. | 28 Nov. |
| 14 | | | (No Class) | *Thanksgiving Break (No Classes)* | *Thanksgiving Break (No Classes)* | *Thanksgiving Break (No Classes)* | |
| Week | 29 Nov. | 30 Nov. | 1 Dec. | 2 Dec. | 3 Dec. | 4 Dec. | 5 Dec. |
| 15 | | | 8.7, 10.1-10.4 | | 10.5, 10.6 | **HW 7** | |
| Week | 6 Dec. | 7 Dec. | 8 Dec. | 9 Dec. | 10 Dec. | 11 Dec. | 12 Dec. |
| 16 | | | 9.2-9.4 | | Review **HW 8** | | |
| Week | 13 Dec. | 14 Dec. | 15 Dec. | 16 Dec. | 17 Dec. | 18 Dec. | 19 Dec. |
| 17 | | | **Final Exam (Ch. 1-7) 3:15–5:15 Witmer 309** | | | | |